Rank-four, eleven-element excluded minors for representability over $\mathbb{F}_5$

Anonymous

2026

Basic concepts

- A *matroid* consists of a finite ground set E and a family $\mathcal{I}$ of *independent sets* that obeys three axioms.
  - $\varnothing \in \mathcal{I}$.
  - Every subset of an independent set is independent.
  - If $I, J \in \mathcal{I}$ and $|I| < |J|$, then some $j \in J \setminus I$ makes $I \cup \{j\}$ independent.
- In a *vector matroid*, the elements label the columns of a matrix. A set is independent exactly when its columns are linearly independent.
- A matroid is $\mathbb{F}$-*representable* if some matrix over the field $\mathbb{F}$ realizes its independent sets in this way.

Bases, duality and minors

- A *basis* is a maximal independent set. All bases have the same size, the *rank*. The ground set and the bases determine the matroid.
- For example, the columns labelled $0, 1, 2$ of $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ give the bases $\{\{1, 2\}, \{0, 2\}\}$.
- The *dual* M^* has the same ground set as M and bases $\{E \setminus B : B \in \mathcal{B}(M)\}$.
- A *minor* is obtained by deletions and contractions.
  - Deletion $M \setminus T$ keeps the independent sets disjoint from T. For a vector matroid, delete the columns in T.
  - Contraction is $M/T = (M^* \setminus T)^*$. For a vector matroid, quotient by the span of the columns in T and keep the remaining columns.

Problem statement

- A matroid is *quinary* if it is representable over $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$.
- An *excluded minor* for quinary representability is a nonquinary matroid all of whose proper minors are quinary.
- It suffices to check M/a and $M \setminus a$ for every $a \in E(M)$, since representability is preserved under taking minors.

Classification

There are exactly **35 isomorphism classes** of rank-four, eleven-element excluded minors for quinary representability.

LLM-assisted computation

- A direct prompt left the classification incomplete. We used an LLM with computational tools to develop and carry out the search.
- The proof reduces the classification to a finite search that enumerates all candidates. SageMath handles exact isomorphism checks, and natural joins combine compatible deletion data.
- We checked separate certificates for nonrepresentability and for the immediate minors of every final matroid.

Proof outline

View an eleven-element excluded minor M as an extension of a ten-element quinary *parent* $N = M \setminus e$.

1. Generate all possible parents N up to isomorphism.
2. For each N, recover all nonquinary extensions M whose deletions are quinary. For $f \in E(N)$, obtain possible $M \setminus f$ by extending $N \setminus f$ by e.
3. Test all single-element contractions of each candidate and remove isomorphic repeats.

Step 1: generating parents

Every parent is connected and simple of rank four. Simplicity rules out loops and parallel pairs.

Matrices over $\mathbb{F}_5$ are *field-isomorphic* if they differ by invertible row operations, nonzero column scalings and a column permutation.

- Any basis can be reduced to I_4, so start with I_4. At each size, append every unused projective point (one nonzero vector per one-dimensional subspace). Use Sage to retain one matrix per field-isomorphism class.
- Convert the ten-column matrices to basis families, deduplicate up to matroid isomorphism and keep the connected parents.
- Keep the matrix catalogue separately. One matroid can have several field-isomorphism classes of representations.

Step 2: labelled extensions

Fix the parent labels $P = \{0, \ldots, 9\}$ and the new label $e = 10$. Let $\mathcal{P}$ contain one labelled representative of each connected simple rank-four quinary matroid on ten elements.

For each $N \in \mathcal{P}$, let $\mathcal{D}(N)$ be the set of simple rank-four matroids M on $P \cup \{e\}$ such that

- $M \setminus e = N$,
- M is nonquinary, and
- $M \setminus a$ is quinary for every $a \in P \cup \{e\}$.

We keep the labels fixed until the final isomorphism checks. Each $N \setminus f$ still has rank four because N is connected.

Step 2: compatibility of deletions

The bases avoiding e are fixed by N. For each triple $T \subseteq P$, let x_T record whether $T \cup \{e\}$ is a basis. Let $\pi_f(x)$ keep the bits for triples avoiding f.

For $Q = N$ or $N \setminus f$, let $\mathcal{R}(Q)$ contain the bit rows on triples of $E(Q)$ for all simple rank-four quinary extensions by e.

- $x \in \mathcal{R}(N)$ gives a quinary extension of N.
- $\pi_f(x) \in \mathcal{R}(N \setminus f)$ gives a quinary deletion at f.

Key fact. If the proposed bases avoiding each element define a simple rank-four matroid, the full family defines one too. Any two bases use at most eight of the eleven elements, so basis exchange can be checked in a common deletion. Each pair of elements also survives in some simple deletion.

$$\mathcal{S}(N) = \{x : \pi_f(x) \in \mathcal{R}(N \setminus f) \text{ for every } f \in P\} \setminus \mathcal{R}(N).$$

These rows correspond bijectively to $\mathcal{D}(N)$.

Step 2.1: labelled representations

For $Q = N$ or $N \setminus f$, a *labelled representation* of Q is a matrix over $\mathbb{F}_5$ whose columns carry the element labels of Q and realize its bases.

Two such matrices are *labelled projectively equivalent* if row operations and nonzero column scalings relate them with labels fixed. We build $\mathcal{L}(Q)$ with one matrix per class.

1. Take one matrix from each field-isomorphism class whose matroid is isomorphic to Q. Align its labels with Q using one matroid isomorphism.
2. Apply the automorphisms of Q to these column labels to reach all other alignments.
3. Normalize the resulting matrices and keep the distinct ones.

Step 2.1: computing automorphisms

Build a coloured bipartite incidence graph for Q.

- Use a blue vertex for each element and a red vertex for each basis.
- Connect an element vertex to a basis vertex exactly when the element belongs to the basis.
- The colour-preserving graph automorphisms induce exactly $\mathrm{Aut}(Q)$ on the element vertices.

Sage supplies generators of this group. Repeatedly apply the generators and keep each new normalized matrix to exhaust the orbit.

Step 2.1: normalizing matrices

Fix an ordered basis $B = (b_1, \ldots, b_r)$ of Q. Reduce its columns to I_r by Gaussian elimination and write the remaining block as Z. The bases of Q determine the support of Z.

$$Z_{ij} \neq 0 \quad \Longleftrightarrow \quad (B \setminus \{b_i\}) \cup \{j\} \in \mathcal{B}(Q).$$

Equivalent reduced blocks differ only by row and column scalings.

- Form the bipartite graph of rows and columns of Z, with an edge for each nonzero entry.
- Fix a spanning forest. In each tree, start with a root multiplier of 1 and choose the remaining multipliers so every tree entry becomes 1.
- The resulting block is unique for the equivalence class. Any further scalings that preserve the tree entries cancel on every edge of that component.

Step 2.2: building the extension tables

For each $V \in \mathcal{L}(Q)$, append each unused point v of $\mathrm{PG}(3,5)$ as column e. The matrix $[V \mid v]$ represents a simple rank-four quinary extension of Q. Its row is

$$z(V,v)_T = \mathbf{1}\{\det[\, V_T \mid v \,] \neq 0\}, \qquad T \in \binom{E(Q)}{3}.$$

The distinct rows obtained this way are exactly $\mathcal{R}(Q)$.

Batching the determinant tests. For each triple, expand along v to write the determinant as $c_T^{\mathsf{T}} v$. Stack the cofactor rows in C and the 156 projective point columns in D. The product CD over $\mathbb{F}_5$ gives all tests for this V at once.

Step 2.3: compressing the tables

Dependent triples always give bit 0. Independent triples spanning the same rank-three flat (hyperplane) H have the same bit y_H, since they span e exactly when H does.

Let $\mathcal{H}(N)$ be the hyperplanes of N. The columns visible in the deletion of f are

$$C_f = \left\{ \mathrm{cl}_N(T) : T \in \binom{P \setminus \{f\}}{3},\ r_N(T) = 3 \right\},$$

where $\mathrm{cl}_N(T)$ is the set of elements spanned by T in N. Compress $\mathcal{R}(N \setminus f)$ to A_f on C_f and $\mathcal{R}(N)$ to A on $\mathcal{H}(N)$. The same solutions in hyperplane coordinates are

$$\widehat{\mathcal{S}}(N) = \{y \in \{0,1\}^{\mathcal{H}(N)} : y|_{C_f} \in A_f \text{ for every } f \in P\} \setminus A.$$

Step 2.4: natural join

- As in a database join, combine two rows exactly when they agree on all shared columns.
- Join the deletion tables A_f one at a time. Every hyperplane H appears in a table, since deleting any $f \notin H$ leaves its spanning triples intact.
- Remove the full rows that appear in A. The remaining rows form $\widehat{\mathcal{S}}(N)$.

$$\widehat{\mathcal{S}}(N) = (A_0 \bowtie \cdots \bowtie A_9) \setminus A.$$

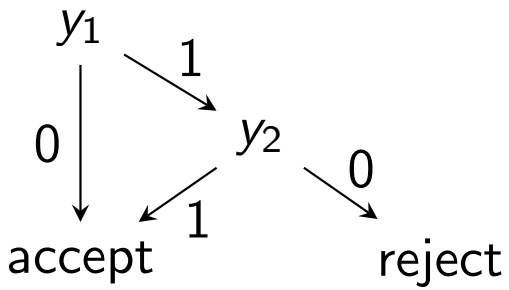
The eleven-element search used this method for every connected parent.

Step 2.4: SAT enumeration

Use one Boolean variable y_H per hyperplane.

1. Encode each A_f as a binary decision diagram and require its root to accept.
2. Block each row of A.
3. Record and block each solution until UNSAT.

Example table $\{00, 01, 11\}$



At a node testing y_H, let v be its value and ℓ, h the values of its $0, 1$ branches. Its equivalence and four clauses are

$$v \leftrightarrow ((\neg y_H \wedge \ell) \vee (y_H \wedge h))$$

$$(\neg v \vee y_H \vee \ell), \qquad (\neg v \vee \neg y_H \vee h),$$
$$(v \vee y_H \vee \neg \ell), \qquad (v \vee \neg y_H \vee \neg h).$$

To block row a, add $\bigvee_{H \in \mathcal{H}(N)} (y_H \neq a_H)$. SAT cross-checked the joins at nine and ten elements.

Step 3: testing contractions

For each $M \in \mathcal{D}(N)$ and each $a \in E(M)$, test M/a for quinary representability.

- Each single-element contraction has rank three because M is simple of rank four.
- Simplify M/a by deleting loops and keeping one element from each parallel class. A representation of $\mathrm{si}(M/a)$ can restore the removed elements as zero columns or parallel copies.
- Compare each simplification with the complete rank-three quinary catalogue generated by the same method as in Step 1.
- Keep the candidates that pass every test and use Sage to deduplicate them up to isomorphism.

Search results and certificates

Stage	Count
Simple ten-element parent classes	58,756
Connected parent classes in $\mathcal{P}$	58,672
Labelled pairs (N, M) with $M \in \mathcal{D}(N)$	659
Classes after contraction tests and deduplication	**35**

We verified each final representative independently.

- A checked DRAT refutation proves the determinant constraints for a representing matrix are unsatisfiable.
- We checked 22 matrices witnessing representability of its deletions and contractions.

A separate incidence-graph check verified pairwise nonisomorphism.

Future work

Complete the classification on eleven elements.

- Duality pairs rank r with rank $11 - r$, so the rank-four classification also gives the rank-seven classification.
- Existing classifications cover ranks up to three. Up to duality, rank five is the remaining case.
- The deletion-compatibility argument still applies at rank five, since $11 > 2 \cdot 5$.

Campbell (2020), *The rank-3 excluded minors for GF(5)-representability*